## □ AI-HAT (V1) — TECHNICAL EXECUTION PLAN

This is structured into:

1. System Architecture
2. Detailed Process Flow
3. UI/UX Plan (Screen-by-screen)
4. Backend & Doctor Dashboard
5. Hardware Integration Plan
6. AI Module Design
7. Data Flow & APIs

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### 🏗 1. SYSTEM ARCHITECTURE (V1)

```
[User]
  ↓
[AI-HAT Device (Tablet + Vitals)]
  ↓
  ├── Edge Layer (UI + AI Triage Lite)
  ├── Integration Layer (Vitals Sync)
  ↓
[Cloud Layer]
  ├── Telemedicine Server (WebRTC)
  ├── Backend APIs
  ├── Database
  ↓
[Doctor Dashboard]
```

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### 🔄 2. DETAILED PROCESS FLOW

#### ◈ MASTER FLOW

```
START
 ↓
User taps "Start Checkup"
 ↓
AI Symptom Intake (Voice/Text)
 ↓
AI generates structured summary + risk level
 ↓
Vitals captured (SpO2, HR, BP)
 ↓
Display combined report
 ↓
```

Prompt: "Consult Doctor?"
↓
IF YES → Initiate Video Call
↓
Doctor views data + interacts
↓
Prescription / Advice
↓
END

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### ◈ AI TRIAGE FLOW (Agent Logic)

Input: Raw user speech/text
↓
Speech-to-Text (if voice)
↓
Symptom Extraction
↓
Keyword + Intent Mapping
↓
Risk Scoring Engine
↓
Output:
- Symptoms list
- Duration (if captured)
- Risk: Low / Medium / High

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### ◈ DOCTOR FLOW

Doctor logs into dashboard
↓
Sees incoming consultation request
↓
Accept call
↓
Views:
- Patient summary
- AI triage output
- Live vitals
↓
Consults via video
↓
Uploads prescription / notes
↓
Session ends

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## 📋 3. UI / UX PLAN (VERY IMPORTANT)

Keep this **ultra-simple**. Your users may be:

- Workers
- Rural population
- Non-tech users

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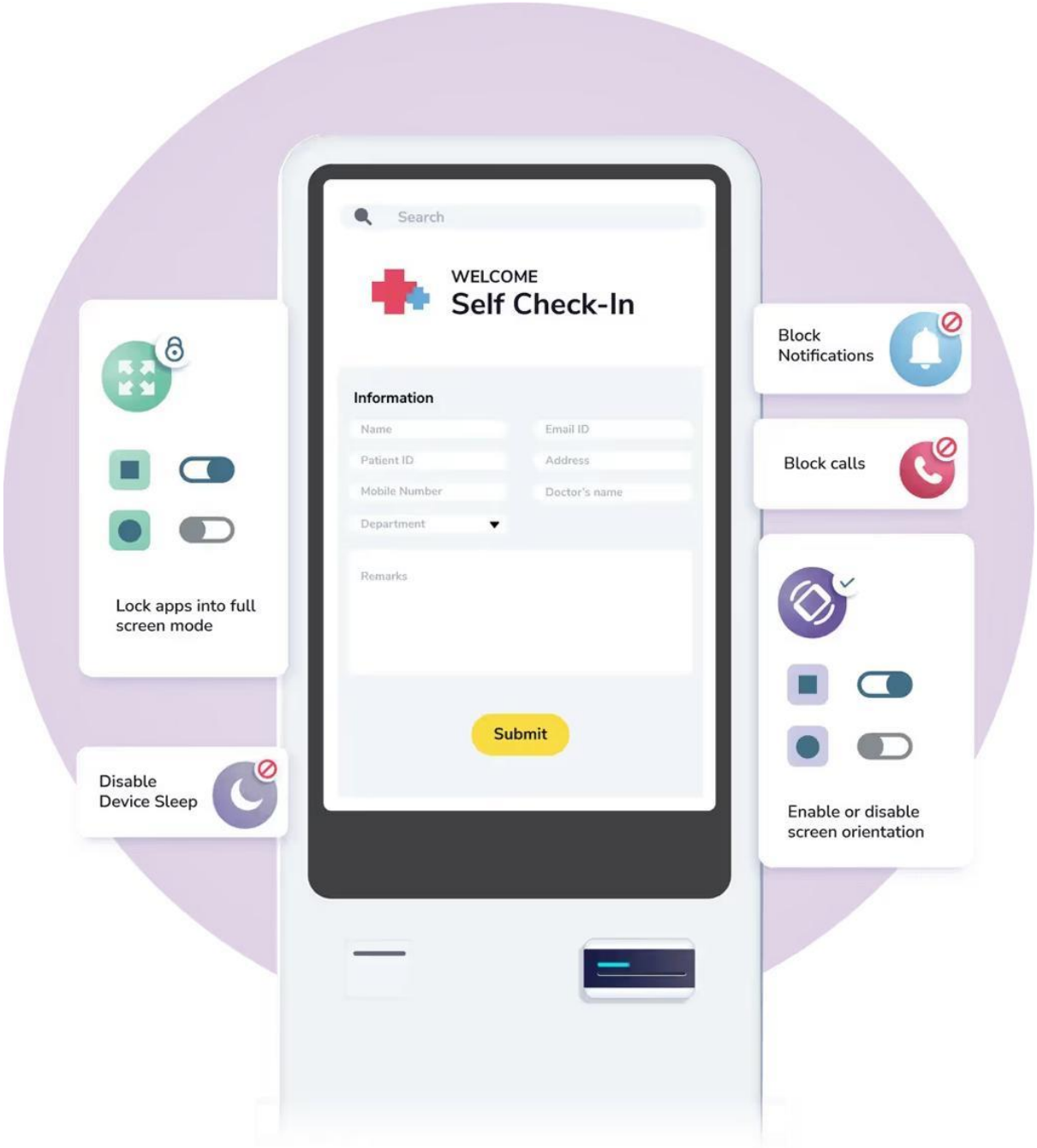
### ◈ SCREEN 1: HOME




6

**Elements:**

- Big Button: **"Start Checkup"**
- Language selection (top corner)
- Optional: Emergency button (future)

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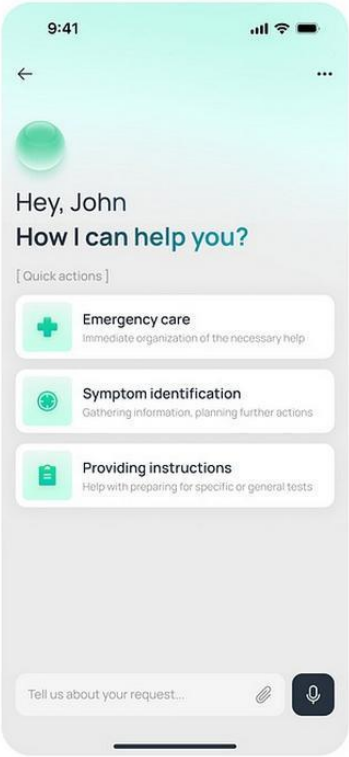
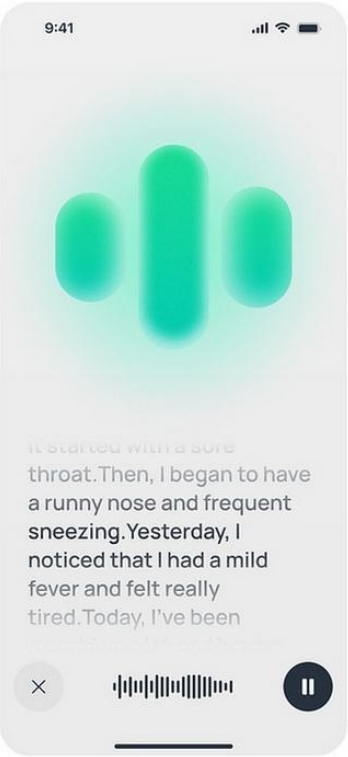
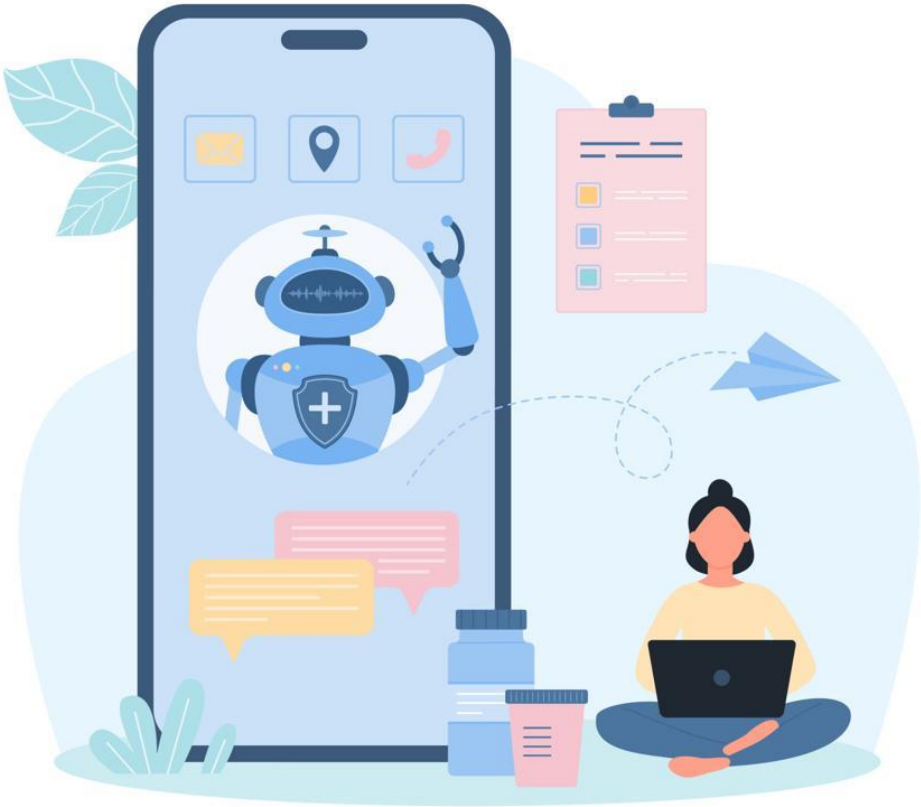
### ◈ SCREEN 2: AI INTERACTION


9:41
Hey, John
How I can help you?
[ Quick actions ]
Emergency care
Immediate organization of the necessary help
Symptom identification
Gathering information, planning further actions
Providing instructions
Help with preparing for specific or general tests
Tell us about your request...


9:41
throat.Then, I began to have
a runny nose and frequent
sneezing.Yesterday, I
noticed that I had a mild
fever and felt really
tired.Today, I've been



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**Elements:**

- “Tell me your problem”
- Mic button 🎤
- Text fallback input
- Chat-style interaction

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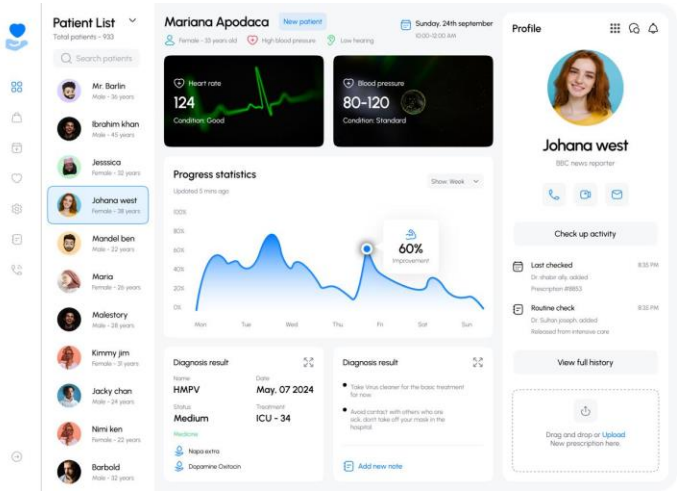
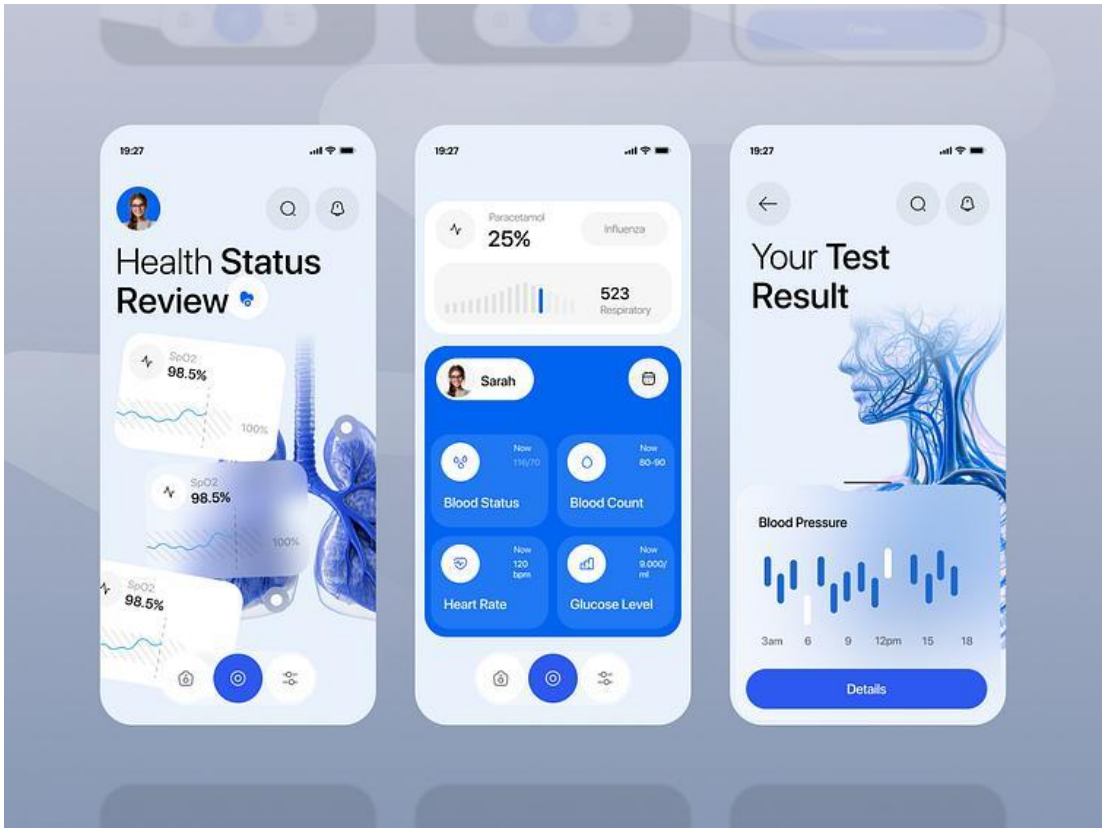
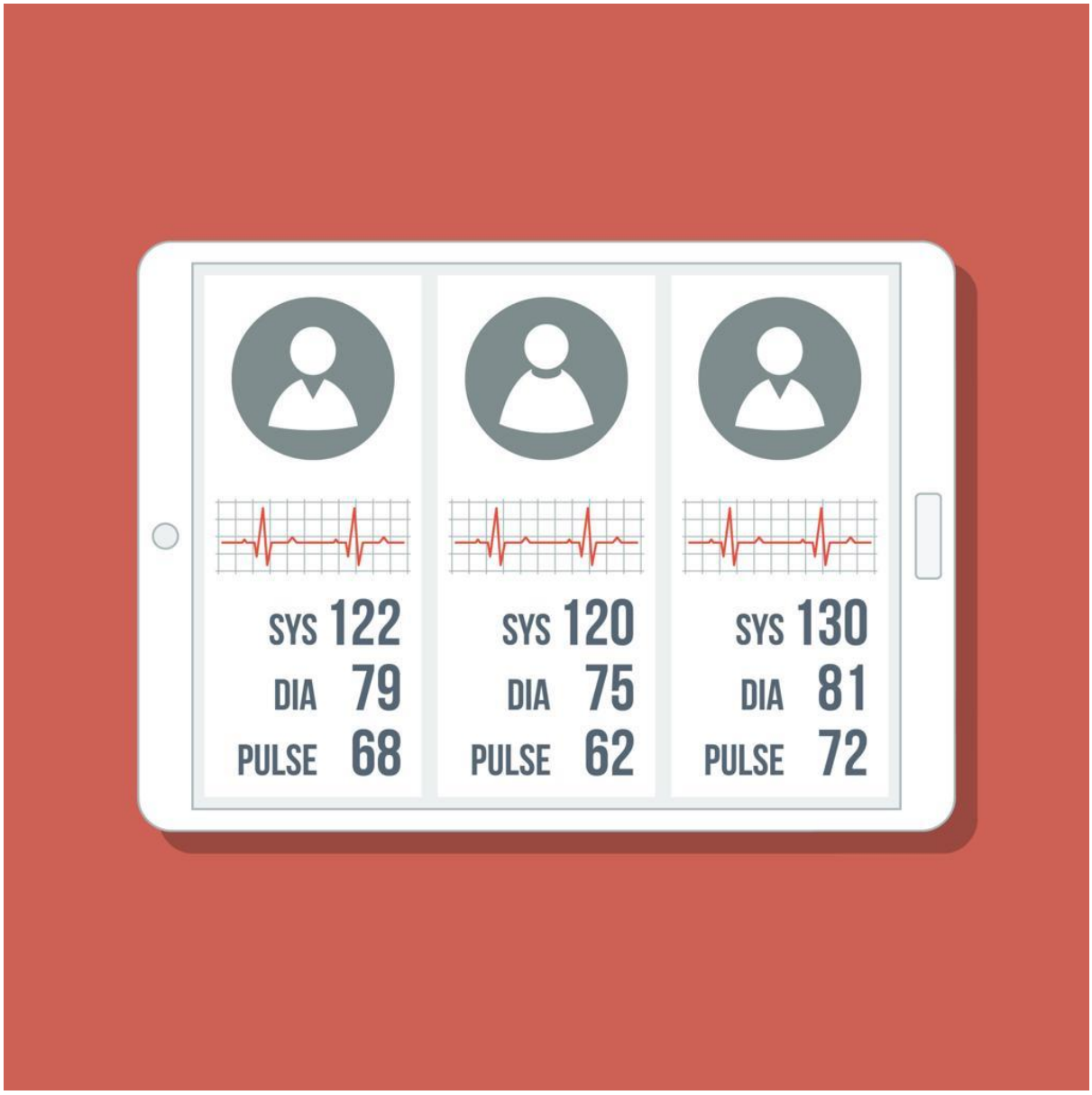
### ◈ SCREEN 3: VITALS DISPLAY



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**Elements:**

- SpO2
- HR
- BP
- Status indicators (Normal / Alert)

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### ◈ SCREEN 4: SUMMARY + DECISION

**Show:**

- AI Summary

- Risk Level (Color-coded)
    - Green / Yellow / Red

**Button:**

👉 **“Connect to Doctor”**

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### ◈ SCREEN 5: VIDEO CONSULTATION

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**Layout:**

- Doctor video (main)
- Patient preview (small)
- Side panel:
    - Vitals
    - AI summary

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### ◈ SCREEN 6: END SCREEN

- “Consultation Completed”
- Option:
  - View prescription
  - Print / Download

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### □□ 4. DOCTOR DASHBOARD (WEB)

**Core Modules:**

**1. Live Queue**

- Incoming patients
- Risk level tags

**2. Consultation Screen**

- Video panel
- Patient info
- AI summary
- Live vitals feed

**3. Prescription Module**

- Upload PDF / type notes

**4. History**

- Patient logs
- Session records

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### 🔌 5. HARDWARE INTEGRATION PLAN

**Device:**

- Android Tablet (preferred for speed)

**Vitals Device:**

- Choose one with:
  - Serial / Bluetooth / API output

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**Integration Flow:**

Vitals Device → Microcontroller (optional) → Tablet App

OR

Vitals Device → Direct Bluetooth → Tablet

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**Data Mapping Example:**

| Parameter | Source | UI |
|---|---|---|
| SpO2 | Sensor | Display |
| HR | Sensor | Display |
| BP | Device | Display |

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## 🤖 6. AI MODULE DESIGN (PRACTICAL)

**Keep it SIMPLE in V1**

**Components:**

- Speech-to-Text (Google / Whisper API)
- Rule-based + lightweight NLP
- Predefined symptom library

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**Example Logic:**

Input:

"Chest pain and sweating"

Output:

Symptoms:
- Chest pain
- Sweating

Risk: HIGH
Recommendation: Immediate consultation

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👉 Avoid:

- Deep diagnosis
- Complex ML initially

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## 🌐 7. BACKEND & API STRUCTURE

**Core APIs:**

**1. Patient Session API**

- Create session
- Store symptoms

**2. Vitals API**

- Push real-time data

**3. Telemedicine API**

- WebRTC signaling

**4. Doctor API**

- Login
- Accept call
- Upload prescription

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**Suggested Stack:**

**Frontend (Device):**

- Flutter / React Native

**Backend:**

- Node.js / FastAPI

**Video:**

- WebRTC (Agora / Twilio)

**DB:**

- PostgreSQL / Firebase